CORIPET RVM FLEET • DEFECT & OUTLIER DETECTION SPECIFICATION

Comprehensive Mathematical Formulations, 8 Certified Anomaly Regimes, Model Governance & Operational Action Protocols

Executive Purpose: This engineering standard establishes the definitive criteria for detecting, classifying, and handling anomalous machine behavior across Coripet's reverse vending machine (RVM) network (1,681+ machines). The system operates on a **Dual-Track Architecture**: (1) **AI Training Governance** to filter out hardware failure artifacts and avoid gradient corruption, and (2) **Operational Maintenance Dispatch** to provide field engineers and retail account managers with actionable diagnostic alerts.

1. DUAL-TRACK ARCHITECTURAL FRAMEWORK

In smart reverse vending networks, anomalies must never be treated simply as 'bad rows' to blindly delete. A sudden collapse from 1,000 to below 400 bottles indicates either a *hardware failure*, a *store staff bin bottleneck*, or a *supermarket promo expiration*. Discarding the machine would lose valuable commercial intelligence, while blindly training on hardware breakdowns distorts neural network predictions across the entire national supermarket network.

Track A: Machine Learning Training Governance	Track B: Coripet Operational & Field Dispatch
• **Loss Masking (Zero-Mask)**: Prolonged breakdown periods (consecutive zeros) are masked out from Huber loss. • **Target Winsorization**: Single-day extreme spikes (>7,500 bottles) are clipped to 6,500 to prevent squared-error gradient explosions. • **Sample Downweighting**: Capacity-jammed and staff-congested days receive sample weight $w = 0.25$. • **Huber Loss δ=1.0**: Provides linear penalty transitions for heavy-tailed operational residuals.	• **Defective Machine Dispatch**: Automatically tags machines with hardware outages for urgent technician maintenance. • **Store Manager Alert**: Identifies full bin choke-points where staff fail to empty compactors twice daily. • **Commercial Optimization**: Identifies chronic underperformers hidden in basements or parking lots for store entrance relocation. • **Expansion Planning**: Identifies Super-Hubs (>3,000 bottles/day) requiring a 2nd RVM unit.

2. COMPREHENSIVE OUTLIER & DEFECT TAXONOMY (8 CERTIFIED REGIMES)

The system enforces 8 distinct mathematical and operational detection rules during data ingestion and model training:

Regime & Code	Mathematical Detection Condition	Root Cause & Profile	AI Training Action	Field Ops Protocol
1. Hardware Defect / Offline `DEFECTIVE_OFFLINE`	Consecutive zeros ≥ 7 days *OR* >20% zero-collection days across last 60 operational days (Mature only).	Compactor motor failure, optical board burnt, power switch turned off by store.	**Zero-Mask**: Exclude zero days from Huber loss calculation.	**P1 Dispatch**: Send technician to replace compactor/sensor.
2. High Volatility Swing `VOLATILITY_SWING`	Base: $V_{30d} \in [400, 800]$ Peak: $V_{7d} \geq 1.5 \times V_{30d}$ (1000+) Drop: Next $V_{7d} \leq 0.65 \times V_{30d}$ (<400) with CV > 0.85 within 60 days.	Retail campaign coupon expired; holiday tourist boom/bust; staff bin fatigue.	**Regime Segment**: Condition on cyclical calendar; evaluate Presenze drop.	**Campaign Audit**: Verify banner promo schedule vs staff bin emptying.
3. Capacity Jam Bottleneck `CAPACITY_BOTTLENECK`	Presenze ≥ 0.80×Median(P) AND Bottiglie ≤ 0.60×Median(B) *OR* Bottles/Visit drops from ≥35 to <15.	Storage bin full by midday; compactor halted; staff fails to replace collection bag.	**Downweight**: Set sample weight w = 0.25 on congested days.	**Store Alert**: Mandate twice-daily bag changes (12:00 & 18:00).
4. Extreme Bulk Spike `EXTREME_SPIKE_EVENT`	Single-day collection > 7,500 bottles (>99.8th percentile) *OR* Z-score > 4.5 vs machine mean.	School recycling contest, restaurant bulk deposit, technician test loop.	**Winsorize**: Cap target label at 6,500 bottles; Huber δ=1.0.	**Commercial Check**: Verify if bulk partner deposited commercial bottles.
5. Chronic Underperformer `CHRONIC_UNDERPERFORMER`	Store Area > 1,500 m² Density > 1,000/km² Median < 250 bottles for ≥60d Residual < -2.5×MAE.	Hidden machine placement (underground parking, behind carts, zero customer visibility).	**Downweight**: Sample weight w = 0.30; prevent lowering baseline expectations.	**Relocation Notice**: Request store manager move machine to main entrance.
6. Super-Hub Overperformer `SUPER_HUB_OUTLIER`	Median volume > 3,000 bottles/day Residual > +2.5×MAE consistently across mature operating months.	Sole recycling point in municipality; local neighborhood green hub.	**Preserve**: Keep in training with full weight w = 1.0 (valid demand).	**Capital Expansion**: Deploy 2nd RVM unit to double store recycling capacity.
7. Zombie Sensor Trickle `SENSOR_DRIFT_TRICKLE`	Daily volume 1 ≤ B ≤ 5 bottles every day with 0 or 1 user visits for ≥14 consecutive operational days.	Optical sensor dust trigger, misaligned laser barrier, or phantom counting loop while store closed.	**Cleanse**: Zero out ghost readings; mask as offline period.	**Hardware Recalibration**: Clean optical lenses & recalibrate laser gate.
8. Schedule Inconsistency `SCHEDULE_ANOMALY`	Active collection > 500 bottles on verified national closing days (Easter, Christmas, closed Sundays).	External outdoor machine accessed while store closed, or internal clock date/time desynchronization.	**Temporal Harmonize**: Verify external machine attribute; resync NTP timestamp.	**Clock Sync**: Push automated NTP time update to machine onboard PC.

3. DEEP DIVE: THE 600 → 1,000+ → <400 VOLATILITY PATTERN

The pattern where an RVM performs at **600 bottles/day**, surges to **1,000+ bottles/day**, and subsequently crashes to **below 400 bottles/day** is the single most informative operational signature in retail recycling. The system runs a multi-stage diagnostic tree to determine the precise root cause:

Diagnostic Test	Evidence Condition	Diagnosed Root Cause	Corrective Action
`Test 1: Footfall Correlation`	PRESENZE (User visits) crashed by ≥50% concurrently with BOTTIGLIE.	**Customer Demand Shock** Supermarket promotional coupon campaign ended, or summer tourist season concluded.	**Marketing**: Retain machine data in training; condition on calendar month features. Inform retailer of promo drop-off.
`Test 2: Capacity Choke Check`	PRESENZE remained steady (≥80%), but BOTTIGLIE collapsed (<400). Bottles/user dropped from 45 to <15.	**Bin Full / Staff Burnout Jam** High volume overwhelmed internal bins; staff delayed emptying compactor; machine sat in 'Out of Service' lock.	**Operations**: Downweight jammed days in training (w=0.25). Mandate scheduled bag emptying every 4 hours during peak retail.
`Test 3: Mechanical Degradation`	Zero-collection hours spiked during prime hours (10:00-12:00, 17:00-19:00) with erratic sensor reads.	**Compactor / Belt Failure** Heavy 1,000+ surge caused mechanical wear or optical sensor misalignment.	**Field Tech**: Dispatch field technician to service crusher teeth, replace drive belt, and clean barcode optics.
`Test 4: Cannibalization Check`	Nearby RVM within 3 km was activated during the surge period.	**Network Cannibalization** A competitor supermarket or new Coripet machine absorbed local recycling volume.	**Spatial ML**: Update nearest_coripet_km and coripet_rvm_count_5km spatial density features.

4. MATHEMATICAL FORMULATIONS & TRAINING INTEGRATION

1. Machine Volatility Metric: To capture non-stationary regime shifts without noise overfitting, we compute the moving Coefficient of Variation (CV) over a 60-day rolling window:

$\mathbf{CV_t = \sigma_{60d} / \mu_{60d}}$, where machines with $\mathbf{CV_t > 0.85}$ are flagged for regime shift inspection.

2. MultiTask Huber Loss Formulation: To shield the neural network from outlier distortion while maintaining sub-bottle precision, the objective function optimizes the Huber loss on log-transformed targets $z = log(1 + y)$:

$\mathbf{L_\delta(z, z_pred) = 0.5(z - z_pred)^2}$ for $|z - z_pred| \le \delta$, and $\delta\mathbf{(|z - z_pred| - 0.5}\delta\mathbf{)}$ for $|z - z_pred| > \delta$ (with $\delta = 1.0$).

3. Loss Weight Masking Formula: Each observation i receives an automated sample weight w_i prior to gradient backpropagation:

- $w_i = 0.0$ if $i \in$ *DEFECTIVE_OFFLINE* (consecutive zeros in mature stores).
- $w_i = 0.25$ if $i \in$ *CAPACITY_BOTTLENECK* (high visits, choked volumes).
- $w_i = 0.30$ if $i \in$ *CHRONIC_UNDERPERFORMER* (obstructed physical placement).
- $w_i = 1.00$ for standard mature operations and verified super-hubs.

5. AUTOMATED PIPELINE EXECUTION & ARTIFACT SPECIFICATION

The outlier detection engine is deployed in two synchronized execution modes:

• **Automated Pre-Training Gate (Monthly at 05:00 AM)**: Integrated directly into scripts/monthly_automated_retrain.py. Before neural network weights are updated, the pipeline scans the active database, applies loss masking, clips extreme spikes, and archives the verified audit snapshot to models/archive/outlier_machines_report.json.

• **Standalone Field Maintenance Auditor (On-Demand CLI)**: Available at any time via python scripts/audit_machine_health.py. Generates an operational spreadsheet data/reports/defected_and_outlier_machines.csv for retail operations and field engineering teams.

Field Name	Data Type	Description & Content Example	Target Audience
`PF`	String (12)	Unique Physical Point ID (e.g., MC069OUNP53U)	All Teams
`Store_Banner`	String	Retail Supermarket Banner (e.g., ESSELUNGA, CONAD)	Account Managers
`Municipality_Region`	String	Geographic location (e.g., Milano (Lombardia))	Field Logistics
`Anomaly_Classification`	Enum	DEFECTIVE_OFFLINE, VOLATILITY_SWING, CAPACITY_BOTTLENECK, etc.	Maintenance Leads
`Downtime_Days`	Integer	Total consecutive days with zero collections (e.g., 14)	Field Engineers
`Median_Actual_Bottles`	Float	Recent 30-day operating median collection volume (e.g., 215.0)	Data Scientists
`Predicted_Expected_Bottles`	Float	AI baseline expected volume based on demographics & store scale (e.g., 850.0)	Commercial Team
`Recommended_Action`	Text	Actionable directive: *'Dispatch technician to unjam compactor motor; check power cable'*	Store Managers & Techs